Outline: 1) Principal bundles Review
2) Connections
   - Parallel transport
   - Gauge gp
3) Curvature
4) ASD connections
5) Y-M equations

E means vector bundle, P means principal bundle

G = Lie gp, X = space

① **Def** A principal G-bundle is a bundle

$$\pi : P \to X$$
$$\circlearrowright$$
$$G$$

where G acts smoothly, $P/G \cong X$ & π is locally $U \times G$.

G is the structure group.

Given $\rho: G \to \mathrm{Diff}(F)$ faithful,
form the <u>associated bundle</u> by

$$\begin{tikzcd} P \times_\rho F := P \times F / G, \arrow[d, "\pi_\rho"'] \\ X \end{tikzcd} \qquad \begin{array}{l} (p,f)\cdot g = (p\cdot g, g^{-1}\cdot f) \\ \pi_\rho([p,f]) = \pi(p) \end{array}$$

Naturally "replaces" fibers of P w/ F.

<u>E.g.,</u> • $G = GL_n\mathbb{R}$ acts naturally on $\mathbb{R}^n$,
so a G-principal bundle $\rightsquigarrow$ vector bundle

• $G = SO(n), SU(n) \rightsquigarrow \mathbb{R}/\mathbb{C}$ v.b.s
w/ fiber/Hermitian
metrics

Reverse this process by taking "frame bundles"

• $G \underset{\rho = Ad}{\curvearrowright} \mathfrak{g} \rightsquigarrow$ "Adjoint bundle" $\mathfrak{g}_P$

2

3 Ways of Thinking About Connections

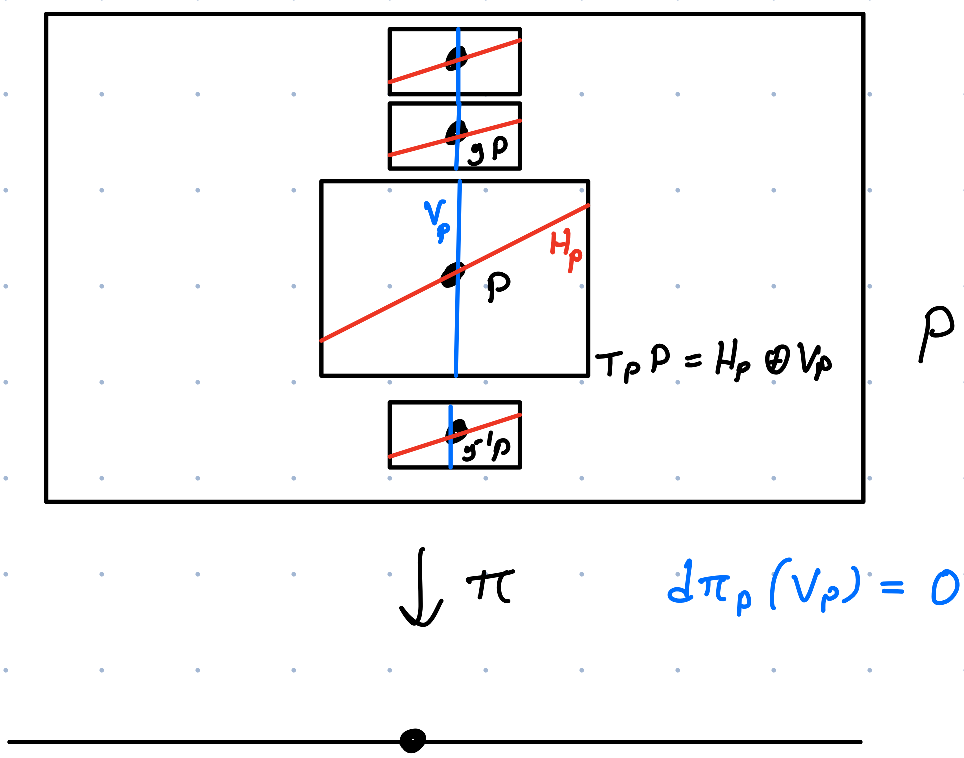
A connection is...

(1) A G-invt splitting $TP = H \oplus \ker(d\pi)$

Tangent bundle of P

Choice of "horizontal" sub-bundle

"Vertical" subbundle



(2) A G-inv't 1-form $A \in \Gamma(\mathrm{Hom}(TP, \mathfrak{g}))$

Go from (2) $\to$ (1) by noting
$\ker d\pi_p \cong \mathfrak{g}$, so $A_p : T_pP \to \ker d\pi_p$.
Let $H_p = \ker(A_p)$.

3) Let $E \to X$ a v.b.. A <u>covariant derivative</u>, i.e. a map

$$\nabla : \Omega^0_X(E) \to \Omega^1_X(E)$$

$\left(\Omega^p_X(E) = \Gamma(\Lambda^p T^* X \otimes E) \right)$

s.t. $\nabla(f \cdot s) = f \nabla s + df \cdot s$

To go from (3) $\to$ (1), take
P = frame bundle of E, $x \in X$
Say $\sigma = (\sigma_1, \dots, \sigma_n) \in \Gamma(P, \underset{x}{\underset{\psi}{U}})$ is

horizontal at x if $\nabla s_i = 0 \ \forall i$. Define
H_p to be $T_p \sigma(X)$ for some horizontal
σ thru p.

E.g. · "Product" connection on $X \times \mathbb{C}^n$
(ordinary differentiation)

· If X a manifold w/ Riemannian metric, $\exists$ canonical connection "Levi-Civita connection"

Denote the space of connections on E $\mathcal{A}_E$

Parallel Transport:

Let $\gamma : [0,1] \to X$ a sm path

$$\rightsquigarrow \quad F_\gamma : P_{\gamma(0)} \to P_{\gamma(1)}$$
$$p \mapsto \tilde{\gamma}(1)$$

$\tilde{\gamma}$ = unique horizontal lift of γ starting at p

$$T_\gamma : E_{\gamma(0)} \to E_{\gamma(1)}$$

Same thing on the frame bundle yields unique linear map

Def $\mathcal{G} := Aut(E)$ is the "gauge group" of E.

$\mathcal{G} \curvearrowright A$ by $u(A) = A - (\nabla_A u) u^{-1}$

as a section of $\mathrm{End}(E)$

or $\nabla_{u(A)} s = u \nabla_A (u^{-1} s)$

Curvature: Define $d_A : \Omega^p_X(E) \to \Omega^{p+1}_X(E)$

by
- $d_A = \nabla_A$, $p = 0$
- $d_A(\omega \wedge \theta) = d_A \omega \wedge \theta + (-1)^p \omega \wedge d_A \theta$

Unlike the de Rham complex, $d_A d_A \neq 0$.

Def $F_A := \nabla_A \nabla_A \in \Omega^2_X(\mathfrak{g}_E)$

Facts
- $F_{u(A)} = u F_A u^{-1}$
- $d_A F_A = 0$. (Bianchi identity).

ASD connections

If X has a metric, we have

$$d_A^* : \Omega_X^{p+1}(E) \to \Omega_X^p(E)$$

w/ $$\int_X \langle d_A \phi, \psi \rangle = \int_X \langle \phi, d_A^* \psi \rangle$$

for forms ϕ, ψ (w/ one having cpt supp)

Recall, there's a map $\star : \Omega_X^k \to \Omega_X^{n-k}$

<u>Hodge Star</u>.

Let X a Riemannian 4-manifold.

Then $\star^2 = 1$ on Ω_X^2, so $\Omega^2_X = \Omega_X^+ \oplus \Omega_X^-$,

the ± 1 eigenspaces of $\star$

This extends to a splitting of bundle-valued forms, write $F_A = F_A^+ \oplus F_A^- \in \Omega_X^+(\mathfrak{g}_E) \oplus \Omega_X^-(\mathfrak{g}_E)$

<u>Def</u> A is <u>ASD</u> if $F_A^+ = 0$.

<u>Fact</u>: Depends only on conformal class of metric.

<u>Context in Yang-Mills theory</u>

<u>Def</u> The <u>Yang-Mills action functional</u> is

$$YM(A) = \|F_A\|_{L^2}^2 = \int_X |F_A|^2 \, d\mu$$

The YM equations are the Euler-Lagrange equations for this functional:

$$d_A^* F_A = 0. \qquad (2^{nd} \text{ order non-linear PDE})$$

ASD connections solve this. They minimize $YM(A)$.

To see this:

<u>Fact</u>: $d_A^* = -\star d_A \star$.

Suppose $F_A^+ = 0$. Then $F_A = F_A^-$, so $\star F_A = -F_A$. Then

$$d_A^* F_A = -\star d_A \star F_A = \star d_A F_A = 0$$

By the Bianchi identity.

Minimality follows from Chern-Weil theory.

YM is conformally invariant <u>only</u> in dimension 4.

Scaling metric by λ scales norm on 2-forms by λ^{-2}, volume form scales by λ^d, so

$$\int |F|^2 \, d\mu \to \int \lambda^{d-4} |F| \, d\mu.$$

Invariant when $d=4$,